

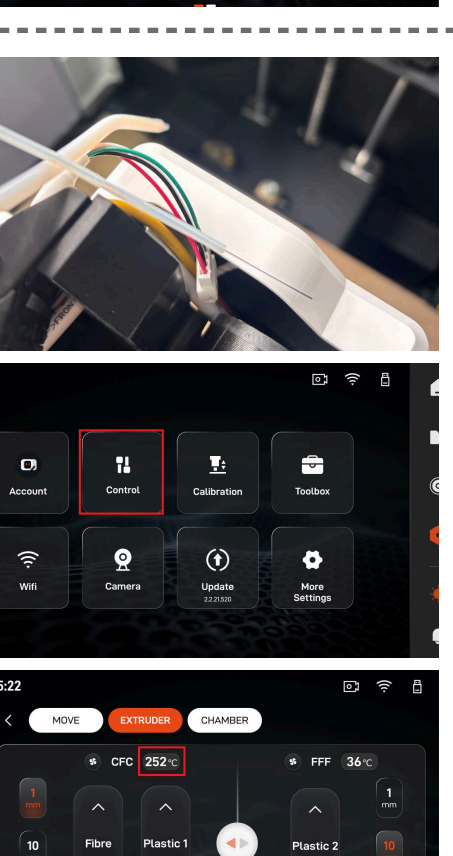
# Fiber Unclogging in Composite Hotend

This SOP describes how to clear fiber clogs in the composite hotend in FibreSeeker 3, aiming to resolve clogging safely and effectively.

## 1. Tools:

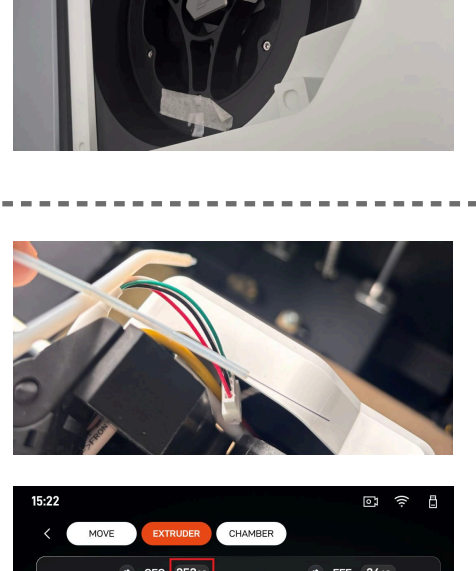
H3, H2.5, H1.5 hex keys, Phillips screwdriver, unclogging pin tool, heat-resistant gloves.

## 2. Precautions:



Troubleshooting requires heating. Wear heat-resistant gloves when working near the print hotend to prevent burns. Before troubleshooting, check the fiber spool for breaks or a rough, gritty feel.

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## Step 1 Heat the Left Nozzle

Heat the left nozzle to the material's printing temperature.

Lower temperatures may block extrusion.



## Step 2.1 Check Extrusion by Removing the PTFE Tube

Remove the PTFE tube. Observe whether the fiber moves.

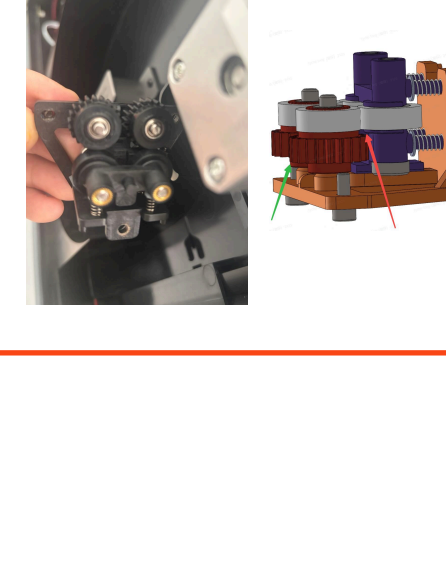
Tap the settings icon on the right side of the toolbar, then select "Control" – "Extruder".

Set the temperature to approximately 250°C and the length to 50mm, then tap the down arrow to extrude.

If fiber comes out of the PTFE tube at the print hotend, skip to Step 11.

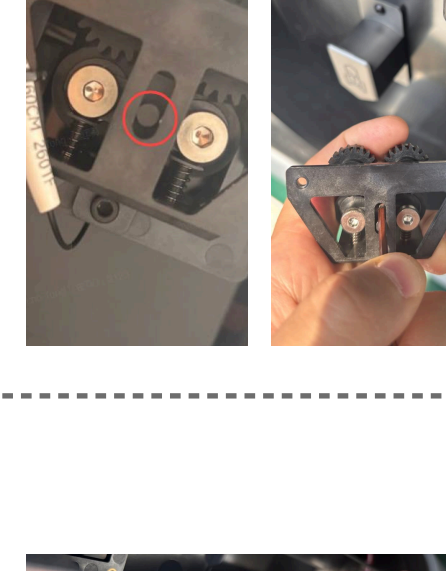


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## Step 2.2 Remove the Fiber

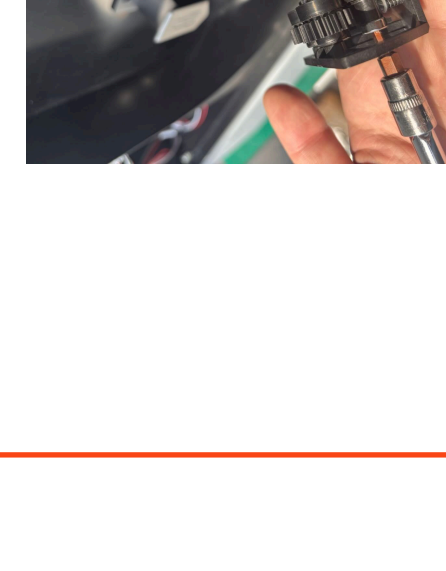
If the fiber does not move, try pulling it out by hand or with tweezers. If it comes out (the fiber broke in the middle), skip to Step 2.4. If it cannot be pulled out, proceed to Step 2.3.



## Step 2.3 Observe the Spool

If the fiber cannot be pulled out, remove the fiber compartment cover. Select 50mm and press extrude twice.

Watch whether the spool rotates. If it does not rotate, go to Step 3. If it rotates, go to Step 2.4.



## Step 2.4 Extrude and Observe

If the spool rotates, extrude 10–12 times until fiber appears inside the PTFE tube within the chamber.

Then reinsert the tube and extrude to see if fiber exits the nozzle. If fiber comes out, skip to Step 10. If the spool rotates but no extrusion occurs and you hear fiber cracking, go to Step 6.

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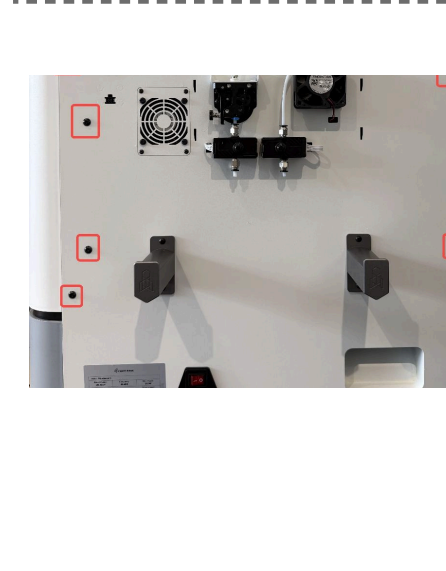
## Step 3 Remove the Side Panel of the Extruder

If the spool does not rotate, inspect the extruder. Use an H3 hex key to remove the 2 screws connecting the rear and side panels and the 4 top fixing screws, then take off the side panel of the extruder.



## Step 4.1 Inspect

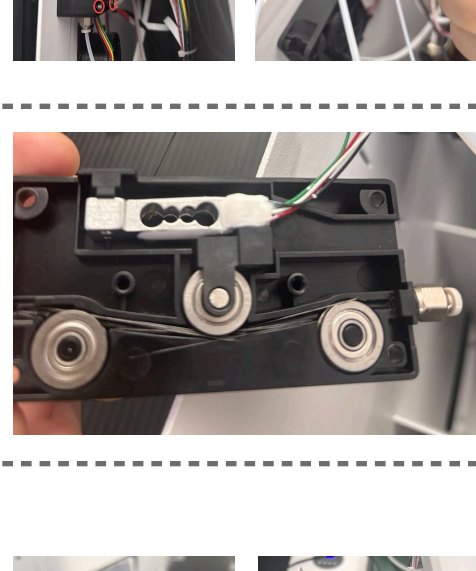
The fiber extruder may have fiber debris (generally visible as fibers sticking out without disassembly).



## Step 4.2 Disassemble

Use an H2.5 hex key to loosen 3 fixing screws and remove the extruder.

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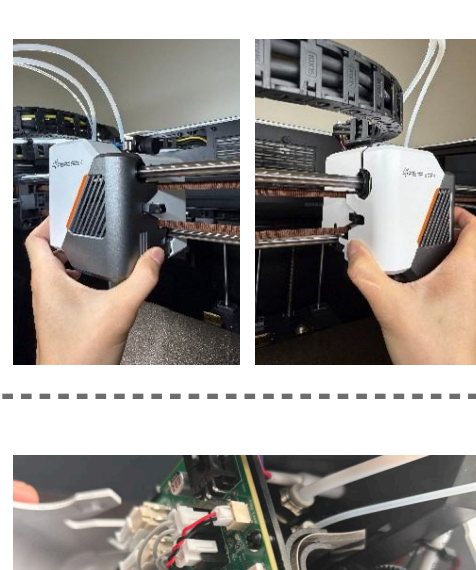
## Step 4.3 Clean

Remove all fiber debris from the extruder and the fiber path.



## Step 4.4 Check

Check the rubber wheel for wear. If severely worn, replace it.



## Step 4.5 Check Extruder Gear Offset

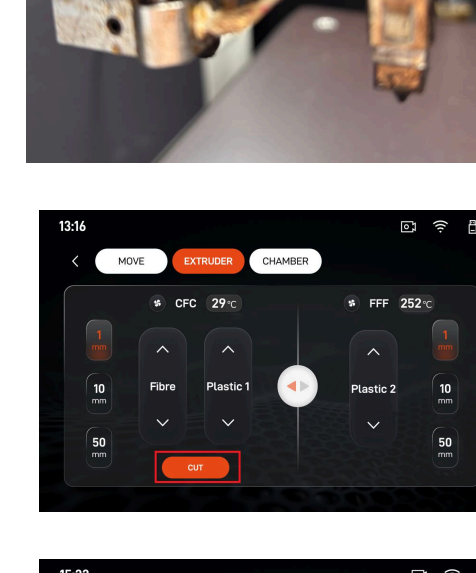
Check whether the extruder gear is offset. While rotating it, observe its movement. (Turn the part indicated by the green arrow and check if the rubber wheel indicated by the red arrow is misaligned.)

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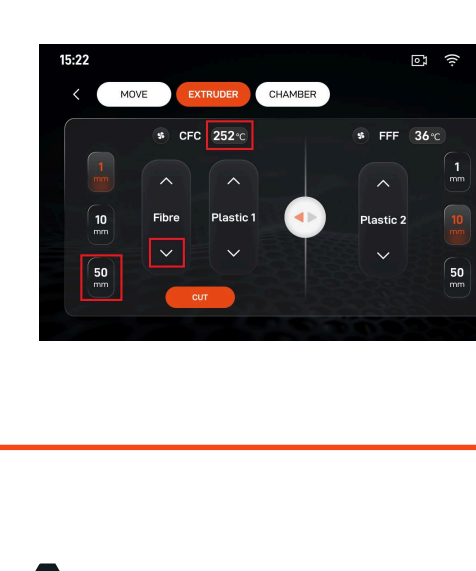
## Step 4.6 Adjust

If offset occurs, press down with a wrench and release immediately to let it spring back.

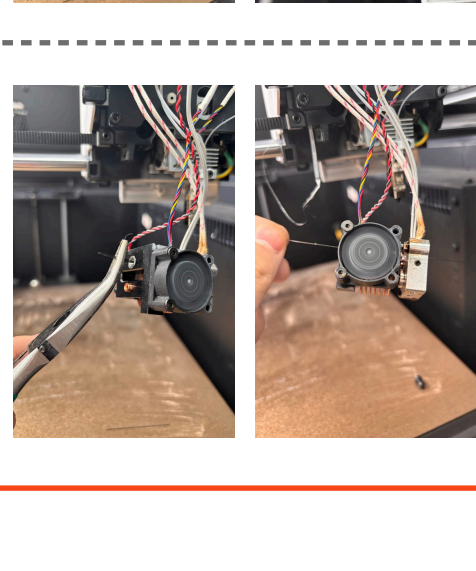


## Step 4.7 Check Adjustment Screws

Use an H3 hex key to inspect the 2 adjustment screws and lightly tighten them by hand. (Tightening the screws increases the pressure on the rubber wheel. Both too tight and too loose will prevent the filament from feeding.)

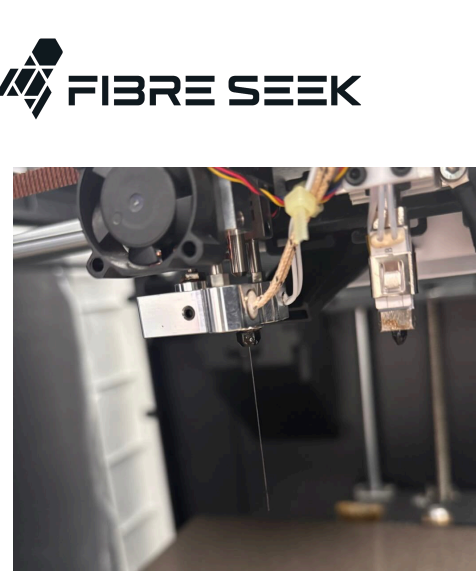


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## Step 5 Reinstall and Test

Reinstall the extruder, reload the filament, and test whether the extruder drives the filament correctly. Continue until filament comes out of the extruder. (After reinstallation, skip to Step 18.)

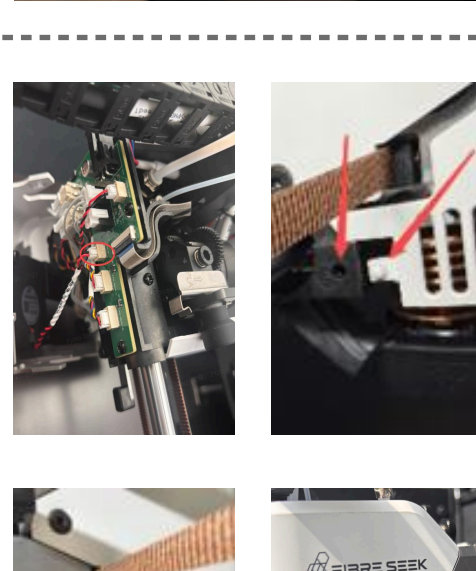


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## Step 6 Remove the Back Panel of the Extruder

The spool rotates with snapping sounds during extrusion. If filament has been extruded more than 12 times without appearing in the PTFE tube, you should also remove the back panel.



## Step 6.1

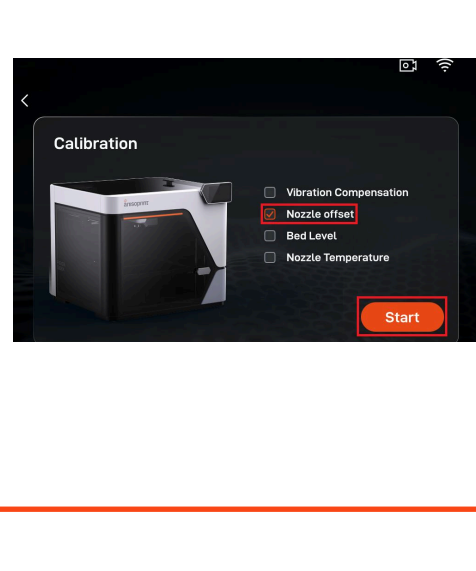
Push the sensor protection cover upward from the bottom to loosen it, then remove it.



## Step 6.2

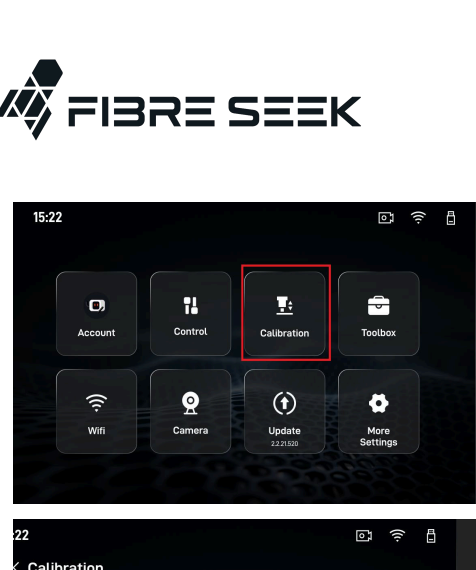
Use an H2.5 hex key to remove the 11 fixing screws on the back cover. Note: The 3 screws at the very top are longer than the others.

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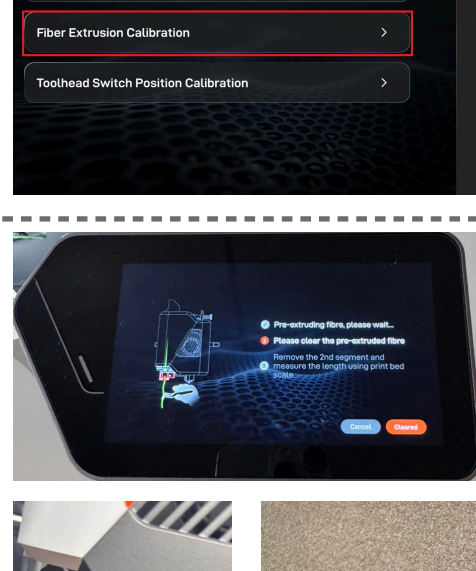
## Step 7 Remove the Tension Sensor

Use an H2.5 hex key to remove the 4 fixing screws securing the tension sensor, then take off the sensor and disconnect the PTFE tube. Then use a Phillips screwdriver to remove the 2 fastening screws.



## Step 8 Inspect

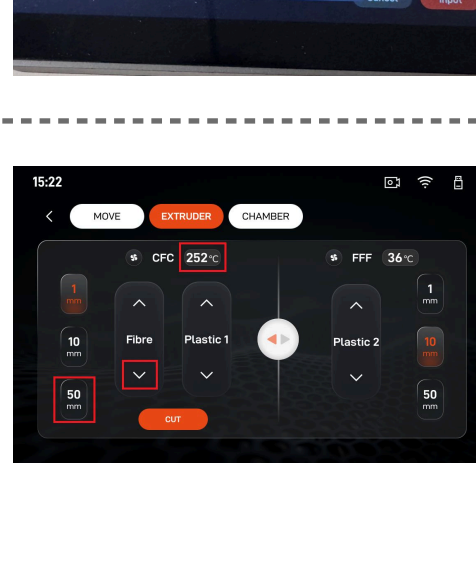
Inspect the inside of the tension sensor for excessive fiber debris. (Internal problems usually cause snapping sounds during extrusion.)



## Step 9 Reinstall, Test, and Secure

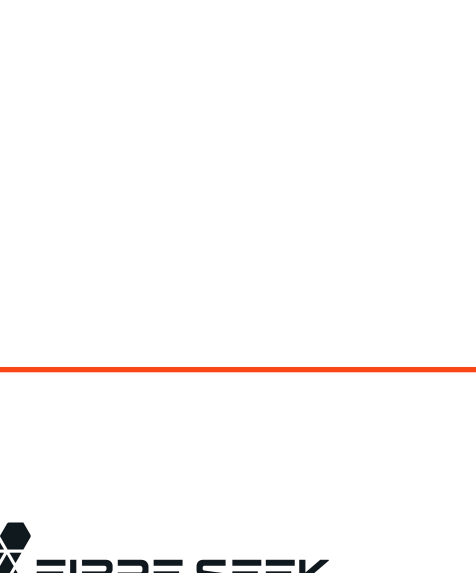
After cleaning, reinstall the sensor. Secure the 2 fastening screws first. Click "Extrude" and check if the filament can pass through the tension sensor. If it can, tighten the 4 fixing screws. (After installation, skip to Step 18.) Then reattach the PTFE tube. Extrude several times. If the filament feeds out from the tension sensor normally, proceed to Step 11.

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## Step 10 Remove the Toolhead Front Cover

Power off the printer. Press the bottom clips on both sides of the front cover simultaneously, then pull the cover forward to remove it.

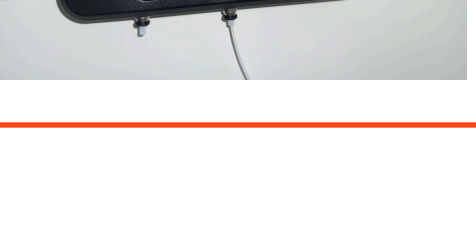


## Step 10.2

Pinch the latch on the fan at the "FAN-R" position on the mainboard and unplug it to remove the front cover of the toolhead.



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## Step 11 Clear Fiber

If the clog is not severe, click "Cut", then use tweezers to pull out any fiber from the guide tube, along with any fiber that comes out with it.

Try extruding 50mm twice and check if fiber comes out of the nozzle. If it does, click "Cut" and continue testing. If still clogged, proceed to the next step.



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## Step 12 Remove the Composite Hotend

Remove the two fixing screws with an H2.5 hex key and take off the composite hotend.



## Step 13 Inspect and Clean

Inspect the fiber path in guide tube and cutter areas for clogs and clean it thoroughly.



## Step 14 Clear the Hotend

Pull out the clogged fiber. Clean the pathway again with an unclogging pin tool, then insert a piece of fiber to ensure it moves through smoothly.



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## Step 15 Verify

Try to extrude 50 mm two to three times. Check if filament comes out of the nozzle. If it does, cut the filament and proceed to the next step.



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## Step 16 Reinstall Assembly

Once cleaned, reinstall the composite hotend assembly in the correct order.



## Step 17 Reinstall the Toolhead Front Cover

Plug the fan cable of the front cover back into the corresponding port on the mainboard. Make sure the two bottom clips align with their holes and snap into place.



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## Step 18 Calibrate

After removing the composite hotend, manually calibrate the nozzle offset. (Only this calibration is needed.)



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## Step 19 Check Fiber Length

### Step 19.1

Click the settings icon on the right side of the toolbar, then select "Calibration" – "Fiber Extrusion Calibration".



### Step 19.2

Follow the on-screen instructions. Measure the length of the second segment, which should be 100mm ± 0.5 mm. (Do not measure the first segment, which is inaccurate.)



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### Step 19.3

Enter the measured length in the Fiber Length field.



### Step 19.4

Then repeat Step 19.1–19.3 twice, ensuring the length from the second extrusion is within 100 mm ± 0.5 mm each time. You may also compare two 100 mm extrusions (the nozzle must be heated above 180°C). If the lengths differ significantly, go back and check Step 4.4.

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## Step 20 Install the Side and Back Panels

Install the 4 fixing screws on top of the side panel. Do not install the two screws that secure the side panel to the back panel yet.



## Step 20.2

Install the 3 longer screws to loosely hold the back cover in place and do not fully tighten. Then install the remaining 8 screws. Align the back cover gaps before tightening all screws evenly.



## Step 20.3

Align the clips on the left and right ends of the sensor cover with the slots in the sheet metal. Once aligned, press the cover downward from the top until it is firmly seated.



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